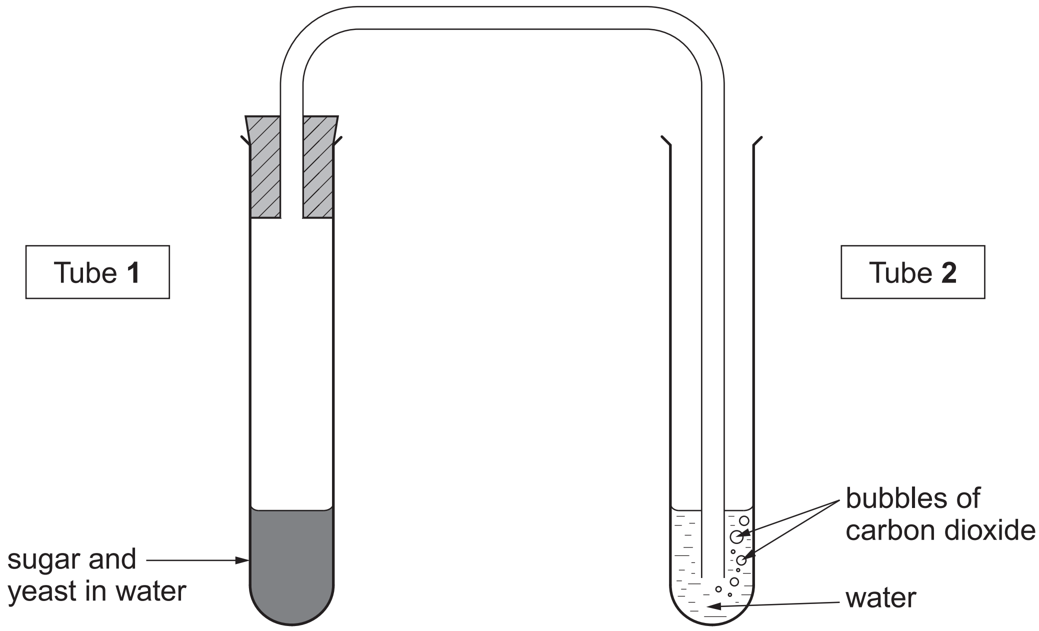
6. Respiration is a series of enzyme controlled reactions which release energy. The word equation for aerobic respiration is shown below.

glucose + oxygen ⟶ carbon dioxide + water + energy

Bethan investigated the effects of temperature on the rate of aerobic respiration in yeast. She used the apparatus below.



Bethan carried out the investigation at a range of temperatures from 15 °C to 45 °C.

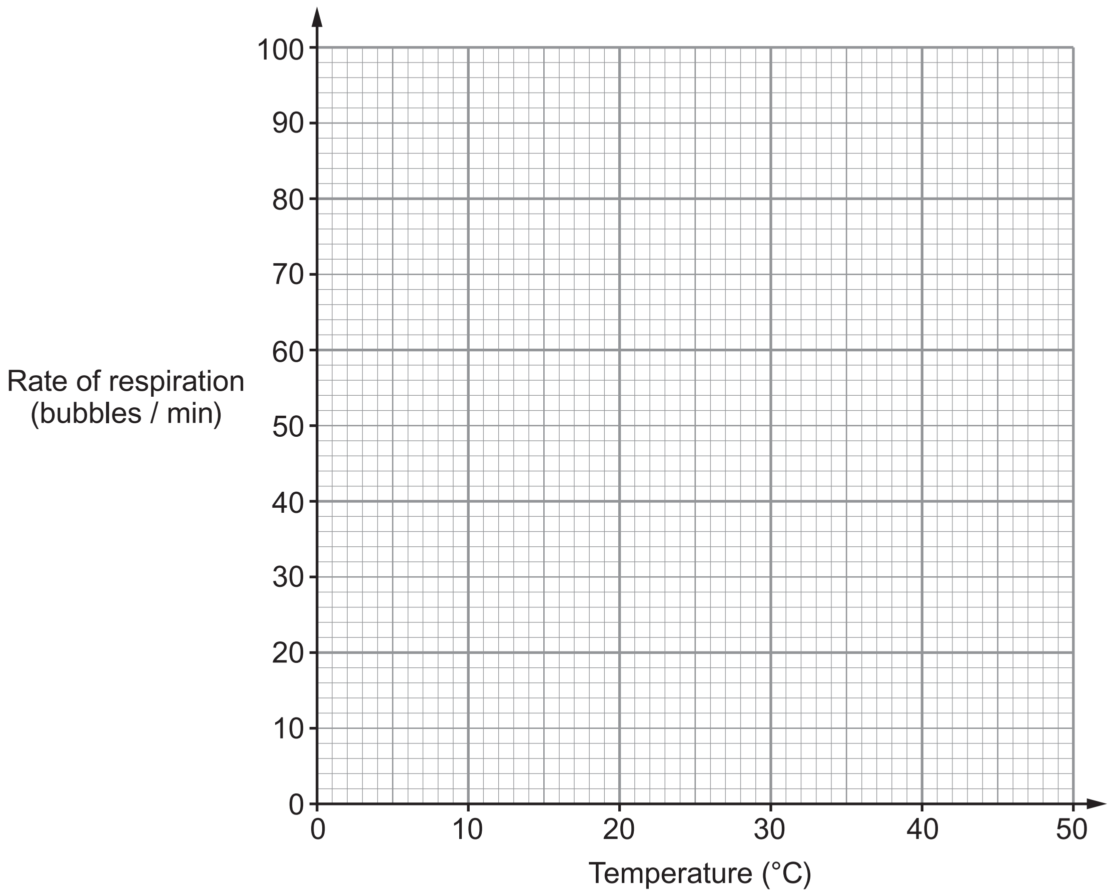
(a) Name the apparatus she could use in order to control the temperature. [1]

....

Examiner only

(b) The table shows the results from the investigation.

Temperature (°C)	Rate of respiration (bubbles / minute)
15	19
25	50
30	64
35	80
40	84
45	67



(i) Produce a graph of the results by: [3]

I. Plotting the values from the table.

II. Drawing a line to join your plots using a ruler.

Examiner only

(ii) I. Describe how the rate of respiration changes as the temperature is increased. [2]

....

....

II. Explain the reason for the change in the rate between 25 °C and 35 °C. [2]

....

....

(c) Bethan was told that she could improve her investigation by confirming that the bubbles of gas given off by the yeast actually were carbon dioxide.

(i) Name the solution she could use in tube **2** of the apparatus to test for carbon dioxide and state the positive result. [2]

....

....

(ii) State **two** factors which she should have kept constant in tube **1** throughout the investigation to ensure that it was a fair test. [2]

1. ....

2. ....

12